2110201: COMP ENG MATH (2024/2) Quiz 6

Name: Student ID: Entry Number:

Part I Choose the correct answer. (5 points, 0.5 points each)

Note: You must answer the following questions in the MCV.

1. If U is an orthogonal matrix, then U is also invertible. (True or False)
2. Given vector $\mathbf{v} = \left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$, $\mathbf{w} = \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$.

   Determine whether these two vectors can form an orthonormal basis. (True or False)

Use the following information for question 3 - 6

Given a basis $\mathbf{B} = \left\langle \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ -2 \end{bmatrix} \right\rangle$.

Let $\mathbf{D}$ be a basis of $\mathbf{R}^2$ and a linear transformation $\mathbf{T}\colon \mathbf{R}^3 \to \mathbf{R}^2$ be represented by a matrix $\mathbf{A}$.

3. Find the matrix that maps vectors from the standard basis to vectors in basis $\mathbf{B}$.

   a) $\mathbf{I}_3$ b) $\mathbf{B}^{\mathrm{T}}$ c) $\mathbf{B}^{-1}$ d) $\mathbf{B}$

4. A vector $\begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$ in basis $\mathbf{B}$ is a vector _____ in the standard basis.

   a) $\begin{bmatrix} 6 \\ 3 \\ -2 \end{bmatrix}$ b) $\begin{bmatrix} 2 \\ 1 \\ -0.5 \end{bmatrix}$ c) $\begin{bmatrix} -2 \\ 3 \\ 6 \end{bmatrix}$ d) $\begin{bmatrix} -0.5 \\ 1 \\ 2 \end{bmatrix}$

5. Find $\mathrm{Rep}_{\mathrm{B},\mathrm{E}_3}(\mathrm{id})$

   a) $\mathbf{I}_3$ b) $\mathbf{B}^{\mathrm{T}}$ c) $\mathbf{B}^{-1}$ d) $\mathbf{B}$

6. Find $\mathrm{Rep}_{\mathrm{D},\mathrm{B}}(\mathrm{T})$

   a) $\mathbf{BAD}^{-1}$ b) $\mathbf{B}^{-1}\mathbf{AD}$ c) $\mathbf{D}^{-1}\mathbf{AB}$ d) $\mathbf{DAB}^{-1}$

7. If $\mathbf{T}\colon \mathbf{R}^2 \to \mathbf{R}^2$ maps (2, 1) to (4, 3), and maps (1, -2) to (-1, 5), what matrix represents T?

   a) $\begin{bmatrix} 7/5 & 6/5 \\ 11/5 & -7/5 \end{bmatrix}$ b) $\begin{bmatrix} 6/5 & 7/5 \\ -7/5 & 11/5 \end{bmatrix}$ c) $\begin{bmatrix} 7/5 & -6/5 \\ 11/5 & 7/5 \end{bmatrix}$ d) $\begin{bmatrix} 7/5 & 6/5 \\ -11/5 & 7/5 \end{bmatrix}$

8. Let $\mathbf{T}$ be a distortion-free linear transformation represented by matrix $\mathbf{A}$.

   Which of the following statements is incorrect?

   a.) $\mathbf{A}$ is an orthogonal matrix. b.) nul($\mathbf{A}$) is trivial.

   c.) $\mathbf{AA}^{\mathrm{T}}$ is an identity matrix. d.) $\mathbf{T}$ is both one-to-one and onto.

Use the following information for question 9 - 10

Let $\mathbf{T}$ be a linear transformation represented by matrix $\mathbf{A}$.

Given that $\mathbf{A}$ is a 2 x 3 matrix with $\mathbf{v_1}, \mathbf{v_2}, \mathbf{v_3}$ as its columns and $\mathbf{A}^{\mathrm{T}}\mathbf{A} = 2025\mathbf{I}$.

9. Find the dot product $\mathbf{v_1} \cdot (\mathbf{v_1} + \mathbf{v_2} + \mathbf{v_3})$

   a.) 0 b.) 45 c.) 90 d.) 2025

10. Which statement(s) are correct?

    (I) $\frac{1}{45}$A is an orthogonal matrix

    (II) $|\det \mathrm{A}| = 45$

    a.) (I) only b.) (II) only c.) (I) and (II) d.) None are correct

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Part II Write solutions and/or give explanations. (5 points)

1. (3 points) Find an orthonormal basis that spans the same subspace as $\left\langle \begin{bmatrix}1\\1\\2\end{bmatrix}, \begin{bmatrix}0\\6\\9\end{bmatrix}, \begin{bmatrix}4\\2\\0\end{bmatrix} \right\rangle$

   by using **Gram-Schmidt algorithm** only.

2. (2 points) Last Friday, one TA was designing a question. They had already written this out:
   - Let $\mathbf{B}$ be a basis of $\mathbf{R}^m$, $\mathbf{D}$ be a basis of $\mathbf{R}^n$, and $\mathbf{P}$ be a basis of $\mathbf{R}^p$
   - Let $h_1\colon \mathbf{R}^m \to \mathbf{R}^n$ and $h_2\colon \mathbf{R}^n \to \mathbf{R}^p$ be linear transformations.
   - Let $\mathbf{A_1} = \mathrm{Rep}_{\mathrm{B},\mathrm{E}_m}(\mathrm{id})$, $\mathbf{A_2} = \mathrm{Rep}_{\mathrm{D},\mathrm{E}_n}(\mathrm{id})$, $\mathbf{A_3} = \mathrm{Rep}_{\mathrm{P},\mathrm{E}_p}(\mathrm{id})$, $\mathbf{A_4} = \mathrm{Rep}_{\mathrm{B},\mathrm{E}_n}(h_1)$, $\mathbf{A_5} = \mathrm{Rep}_{\mathrm{D},\mathrm{E}_p}(h_2)$

   "Find the value of …"

   Then, the earthquake hit. This forced the TA to leave their belongings behind, not able to complete the question. After some time, another TA saw this question and asked: "Why don't we just made the question $\mathrm{Rep}_{\mathrm{B},\mathrm{P}}(h_2 \circ h_1)$? **It should be possible with only the product of some $\mathbf{A_i}$ and/or their inverses.**"

   Is this claim true? If yes, show how to write the product in every single way possible. If not, explain why.